School of Engineering and Applied Sciences

Thesis Proposal

PHYSICS-INFORMED DATA-DRIVEN MODELING FOR HIGH-FIDELITY DIGITAL TWINS OF SERIAL MANIPULATORS

Abstract

Accurate dynamic modeling of serial manipulators is essential for control design, simulation, and the development of high-fidelity digital twins. Traditional physics-based methods, such as the Newton–Euler and Euler–Lagrange formulations, offer interpretability and theoretical foundation but often struggle with complex and nonlinear systems. This thesis proposes a physics-informed data-driven framework to address these challenges by leveraging Sparse Identification of Nonlinear Dynamics (SINDy) and machine learning techniques. The approach automates model generation directly from Unified Robot Description Format (URDF) files, identifying dominant dynamic terms and friction components while integrating physical constraints to ensure interpretability and consistency. The proposed methodology streamlines the modeling pipeline, improves adaptability to changing configurations, and enhances robustness against noise. Validation will be performed using simulated and real-world robotic platforms, with the goal of developing accurate, adaptable digital twins that support advanced control and predictive maintenance in complex robotic systems.

Literature Review

1. Dynamic Models in Robotics: Fundamentals and Applications

Dynamic models play a foundational role in robotics, underpinning key functions such as control system design, simulation, and behavioral analysis. These models are critical for predicting how a robotic system will respond to various inputs and environmental conditions, enabling the development of effective control strategies that ensure precise and stable operation. Beyond motion control, dynamic models also provide vital information regarding joint reaction forces and moments. This data is indispensable for the proper design and selection of mechanical components such as links, bearings, and actuators. Accurately estimating these forces ensures that each component is appropriately dimensioned to endure operational stresses, thereby enhancing the overall reliability and durability of the robotic system [1].

Classical modeling techniques, including the Newton-Euler and Euler-Lagrange formulations, are widely used to derive dynamic equations based on physical principles. While th methods offer a

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structured and theoretically sound approach, their accuracy often hinges on simplifying assumptions that can limit their applicability—particularly for robots with complex architectures or multiple degrees of freedom. In dynamic or uncertain environments, such simplifications may lead to significant modeling inaccuracies, highlighting the need for more adaptable and robust alternatives [2].

A major challenge in traditional modeling is parameter identification, especially in systems governed by nonlinear dynamics. A precise knowledge of physical properties such as mass, inertia, and friction coefficients are essential for constructing reliable models. However, obtaining accurate values for these parameters can be difficult, particularly in robots with intricate mechanical designs or those operating in variable conditions.

To address these limitations, researchers have increasingly turned to data-driven modeling approaches, which allow dynamic models to be constructed directly from experimental or simulation data. These techniques reduce the need for manual derivation and relax the reliance on restrictive assumptions,

making them especially valuable for complex or high-dimensional systems where conventional methods fall short. By leveraging real-world data, these approaches enable the creation of more accurate, adaptable, and generalizable models that can evolve with the system they represent [1].

2. Traditional Methods for Obtaining Dynamic Models

2.1 Newton-Euler and Euler-Lagrange Formulations

The Newton-Euler and Euler-Lagrange formulations are classical tools for deriving dynamic equations of motion in robotic systems. These methods rely on conservation laws and kinematic descriptions, often encoded using Denavit-Hartenberg (DH) parameters [3]. While these analytical approaches offer interpretability and theoretical guarantees, they become increasingly error-prone for robots with many degrees of freedom or nonlinear dynamics [2].

Recursive algorithms have been proposed to improve computational efficiency, especially for closed loop and underactuated systems [2]. Despite these advancements, inaccuracies in parameter estimation and sensitivity to initial conditions often undermine model fidelity, particularly in industrial applications involving variable payloads or changing environments [1].

2.2 Multi-Body Dynamics Simulators

Tools such as Gazebo, Simscape Multibody, and MSC ADAMS are widely used for simulating robotic dynamics using URDF files or CAD-based geometries. These platforms allow v[illegible] prototyping and

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control strategy testing, offering high visual and functional fidelity. However, most simulators implement linearized or simplified versions of the true dynamics, limiting their applicability in highly nonlinear systems [3].

While these simulators support rapid development and testing, they do not inherently enable dynamic model identification or adaptation to changing conditions [9]. Therefore, they are best used in conjunction with experimental calibration and data-driven refinement rather than as standalone modeling solutions [9].

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3. Parameter Identification Techniques

Accurate knowledge of physical parameters—such as mass, inertia tensor, and friction coefficients—is essential for building reliable dynamic models. Parameter identification involves estimating these values from input-output data collected during specific excitation trajectories designed to maximize information content [4].

Classical identification methods often employ least-squares optimization and trajectory planning to estimate inertial properties. However, their success depends heavily on the quality of the base model derived via analytical means [4]. Recent work has emphasized the geometric structure of inertia properties, highlighting configuration-dependent effects that significantly influence dynamic behavior [9]. Incorporating these insights improves both model fidelity and robustness, particularly in articulated systems such as serial manipulators [9].

Additionally, Bayesian regression techniques have been explored to enhance uncertainty quantification and improve convergence in noisy datasets [7].

4. Emergence of Data-Driven Modeling Techniques

In response to the limitations of traditional modeling methodologies, data-driven approaches have gained prominence for their ability to learn complex relationships directly from sensor or simulation data, reducing reliance on manual derivation and simplifying assumptions [8].

These techniques offer several advantages:

- They can bypass symbolic derivation, streamlining the modeling pipeline.
- They handle nonlinearities and unmodeled dynamics more effectively.
- They enable continuous updating of models as new data becomes available.

Among various data-driven methodologies, Sparse Identification of Nonlinear Dynamics (SINDy) has shown promise due to its ability to produce interpretable and compact models directly from time-series observations [8].

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5. Sparse Regression and Its Role in Dynamic Modeling

Sparse regression has emerged as a powerful framework for discovering governing equations from data, particularly in robotics where interpretability, efficiency, and generalization are critical [8]. Unlike black-box machine learning models, sparse regression techniques aim to identify only the fewest terms necessary to describe a system's dynamic producing models that are both accurate and interpretable.

5.1 Introduction to SINDy

One of the most prominent methods is Sparse Identification of Nonlinear Dynamics (SINDy) , introduced by Brunton et al. (2016) [8]. SINDy operates under the assumption that many physical systems can be described using only a small number of active terms from a large library of candidate functions.

This method constructs a dictionary of potential nonlinear basis functions—such as polynomials, trigonometric terms, or interaction terms between variables—and then uses sparse regression algorithms (e.g., Sequential Thresholder Least Squares or LASSO) to select those that best describe the observed dynamics from time-series data [8].

For robotic manipulators, this approach is especially valuable. Traditional dynamic modeling of serial manipulators often relies on Newton-Euler or Euler-Lagrange equations derived manually or symbolically from kinematic parameters [3]. However, these derivations become increasingly error-prone for robots with multiple DOFs, especially when unmodeled effects such as joint friction, backlash, or payload variations are present [4].

SINDy offers an alternative by learning the governing equations directly from input-output data, by passing the need for explicit derivation while still retaining model structure and physical plausibility [8][4].

5.2 Advantages of Sparse Regression in Robotics

1. Model Interpretability

A major strength of sparse regression lies in its ability to deliver interpretable models. By selecting only, the most relevant terms from a predefined functional space, SINDy recovers dynamic equations that resemble classical physics-based formulations, making them easier to analyze, validate, and integrate into control frameworks [8]. This bridges the gap between empirical learning and physics

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based modeling—an essential requirement for safety-critical and high-precision applications such as industrial automation, surgical robotics, and aerospace systems [5].

2. Noise Robustness

Despite working with noisy experimental or simulation data, SINDy demonstrates robustness due to its sparsity-promoting nature. Regularization techniques embedded within the algorithm help prevent overfitting and suppress irrelevant or redundant terms [4]. Omar et al. (2024) showed that SINDy could converge to meaningful solutions even in the presence of significant noise levels, replicating torque curves calculated via closed-form analytical models [4].

3. Direct Learning from URDF Descriptions and Simulation Data

Recent work has explored integrating sparse regression techniques with robotic simulators and Unified Robot Description Format (URDF) files to automate dynamic model discovery [3]. A promising approach involves parsing URDF descriptions to extract inertial and kinematic properties, generating synthetic motion data through simulation, and applying SINDy to discover underlying equations of motion [3][4]. This pipeline eliminates the dependency on manual derivation or parameter identification, streamlining the modeling process for modular or reconfigurable robots [3].

4. Generalization Across Configurations and Payloads

Sparse regression models trained on diverse datasets can generalize across different payloads, link configurations, and operating environments. Uttley and Lee (2020) extended sparse regression concepts using Bayesian inference to improve uncertainty quantification, enabling models to adapt dynamically to changing conditions [7]. This line of research supports the broader goal of creating adaptive digital twins—virtual replicas of physical robots that evolve alongside their real counterparts through continuous data assimilation [5].

5. Integration with Physics Constraints

To ensure physically consistent predictions, researchers have explored hybrid approaches that embed domain knowledge into the SINDy framework. For instance, physics-informed neural networks (PINNs) enforce known conservation laws or boundary conditions during training [5]. Similarly, structured dictionaries built from La grangian mechanics or geometric constraints can guide sparse regression toward discovering valid equations of motion [2][9].

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5.3 Challenges and Future Directions

While sparse regression holds great promise, several challenges remain. One key limitation is the dependence on the quality of the candidate function library. If the true dynamics involve nonlinearities not included in the library (e.g., hysteresis, higher-order interactions), SINDy may fail to capture them accurately [8]. Additionally, current implementations assume uniform sampling and smooth trajectories, which may not always reflect real-world robot behavior [4].

Future work should focus on expanding the dictionary of basic functions to include generalized forces, external torques, and non-smooth dynamics such as contact forces and Coulomb friction [1][9]. Incorporating symbolic regression and automated term generation can further improve flexibility and accuracy [8]. Moreover, integrating sparse regression with online learning architectures would allow for continual model refinement based on real-time sensor feedback—an essential step toward fully autonomous robotic systems.

6. Advantages of Data-Driven Techniques Over Traditional Methods

6.1 Handling Complexity and Nonlinearities

Data-driven methods can effectively capture complex robot dynamics without requiring explicit mathematical models. This is particularly advantageous for robots with many degrees of freedom or complex geometries, where traditional modeling approaches can become unwieldy [3]. These techniques can handle nonlinearities and uncertainties that are difficult to model using traditional approaches, leading to more accurate and robust models [3].

Serial manipulators often exhibit highly nonlinear dynamic behavior due to joint coupling, actuator dynamics, and varying payloads. Traditional methods struggle to account for these effects comprehensively, especially when real-time adaptability is required [1]. In contrast, data-driven models learn directly from operational data and can adaptively capture these intricate interactions without requiring symbolic derivation [3].

6.2 Reduced Modeling Effort

Data-driven approaches significantly reduce the labor-intensive process of symbolic derivation and kinematic analysis [3]. For example, learning dynamics directly from URDF files eliminates the need for manual DH parameter extraction—a step that is time-consuming and error-prone [3]. This capability streamlines the modeling pipeline, allowing engineers to focus on high-level tasks such as controller design and system optimization.

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Additionally, several studies have demonstrated effective modeling pipelines that integrate URDF parsing with data-driven learning, enabling automatic generation of dynamic models tailored to specific robot configurations [4].

6.3 Improved Accuracy and Robustness

Data-driven models can achieve superior accuracy compared to traditional counterparts by compensating for modeling errors and uncertainties inherent in analytical approaches [3]. For instance, SINDy-based models have demonstrated robustness to noisy data by identifying dominant dynamic components and filtering out irrelevant or spurious terms [4].

Regularization techniques such as LASSO and ridge regression further help prevent overfitting, ensuring stable and reliable performance on unseen data [4]. Experimental validation has shown that certain data-driven methods can replicate torque curves calculated via closed-form solutions even at high noise levels, underscoring their practical utility [4].

6.4 Generalization Across Environments and Tasks

A model's ability to generalize across different environments and tasks is crucial for real-world deployment. Traditional models often fail in this regard due to their dependence on fixed assumptions about system behavior. In contrast, data-driven approaches, particularly those leveraging transfer learning and domain adaptation—can be fine-tuned or adapted to new environments with minimal retraining [1].

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7. Conclusion

The evolution of dynamic modeling in robotics highlights a transition from rigid, physics-based formulations to flexible, data-driven paradigms. While traditional methods remain foundational for their interpretability and theoretical rigor, they are increasingly being complemented—or replaced—by data-driven techniques that offer greater adaptability, reduced modeling effort, and enhanced accuracy.

This synthesis of physics-based and data-driven modeling reflects a growing trend in modern robotics research. Frameworks such as SINDy, PINNs, and Bayesian regression offer powerful tools for discovering interpretable models from data while respecting known physical laws. Meanwhile, advances in URDF parsing, parameter identification, and robust model validation continue to bridge the gap between simulation and reality.

As robotic systems grow in complexity and autonomy, the need for high-fidelity digital twins becomes paramount. The fusion of physics-informed and data-driven methods offers a promising path forward—one that can support next-generation robotics in manufacturing, healthcare, aerospace, and beyond.

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Methodology

1. Kinematic and Dynamic Feature Extraction

The modeling process begins with extracting the robot's physical and structural properties from simulation platforms. This includes:

- Parsing the URDF or physical description to retrieve:
  Link lengths, masses, inertias.
  Joint types and configurations.
  Transformation matrices and Denavit-Hartenberg parameters.
- Deriving the full kinematic chain to form the baseline structure needed for dynamic modeling.

2. Dominant Term Identification Using Sparse Regression

To reduce model complexity while preserving accuracy, the SINDy (Sparse Identification of Nonlinear Dynamical Systems) algorithm is applied. This helps:

- Identify the dominant dynamic terms governing the manipulator's motion (inertial, gravitational effects).
- Automatically construct interpretable equations by selecting the most influential terms from a library of candidate nonlinearities.

3. Model Selection via Quadratic Optimization and AIC

The identified models are further refined using optimization and statistical selection criteria:

- A quadratic optimization problem is formulated to estimate the coefficients of the selected terms while minimizing errors.
- Akaike Information Criterion (AIC) is employed to compare competing models, striking a balance between model accuracy and complexity.
- The result is a sparse expressive model that generalizes well and avoids overfitting.

4. Friction Term Identification

Non-idealities like joint friction, often neglected in purely analytical models, are explicitly captured:

- Friction is modeled as an additional term.
- These terms are identified from residuals or learned directly during sparse regression.

Including the friction component enhances model realism and control fidelity.

5. Confidence Estimation Using AI Models

To evaluate the robustness and reliability of the identified model terms:

- A supervised AI model (neural network or decision tree) is trained to estimate the confidence percentage for each selected term.
- This AI layer learns from validation datasets and simulates scenarios to quantify the stability and generalization of model terms across different operating conditions.

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Thesis Proposal

Work/Time Plan

Task	2025				2026								
	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sep
Literature Review													
Development of the proposed approach													
Writing first paper draft													
Submit first paper													
Finalizing the proposed approach													
Submit second paper													
Write the dissertation													
Defend the thesis													

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Expected outcomes:

This research proposes applying data-driven dynamic modeling techniques to extract dynamic models from Unified Robot Description Format (URDF) files, commonly used in robotic simulations. The key objectives include:

- **Automatic Extraction of Kinematic and Dynamic Features from URDF Files:**
  - Parsing URDF descriptions to obtain mass, inertia, and joint parameters.
  - Converting this information into a structured dataset for model training.
- **Applying Data-Driven Techniques to Estimate Dynamic Equations:**
  - Using SINDy to identify dominant terms governing robot motion.
  - Exploring machine learning techniques to incorporate constraints.
- **Validation through Simulated and Real-World Experiments:**
  - Testing on benchmark robotic systems (e.g., KUKA, Stanford Manipulator).
  - Comparing results with traditional Euler-Lagrange models.

Expected Contributions:

This research aims to:

- Develop a framework for generating dynamic models from URDF descriptions.
- Improve accuracy and computational efficiency of robotic dynamics modeling.
- Facilitate enhanced control strategies and adaptive behaviors in robotic manipulators.
- Expand applications in robotic simulation, digital twins, and automated controllers.

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